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DETAIL SPECIFICATION

FLAME ARRESTER, VENTILATION-EXHAUST (NAVAL SHIPBOARD USE)

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers flame arresters for preventing the propagation of flame through exhaust ventilating ducts which may contain flammable mixtures. The flame arresters are for naval shipboard use.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-901	-	Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for
MIL-PRF-16552	-	Filters, Air Environmental Control System, Cleanable, Impingement (High Velocity Type)
MIL-DTL-24441/29	-	Paint, Epoxy-Polyamide, Green Primer, Formula 150, Type IV

(Copies of these documents are available online at https://quicksearch.dla.mil.)

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) DRAWINGS

804-1231181 - Flame Arrester and Filter Assembly

(Copies of this document are available from the applicable repositories listed in S0005-AE-PRO-010/EDM, which can be obtained online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/. Copies of this document may also be obtained from the Naval Ships Engineering Drawing Repository (NSEDR) online at https://199.208.213.105/webjedmics/index.jsp. To request an NSEDR account for drawing access, send an email to NNSY_JEDMICS_NSEDR_HELP_DESK@navy.mil.)

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

S9074-AR-GIB-010/278 - Requirements for Fabrication Welding and Inspection, and Casting Inspection and Repair for Machinery, Piping, and Pressure Vessels

(Copies of this document are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/ by searching for the document number without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. This document is available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM A123/A123M - Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products

ASTM A653/A653M - Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process

ASTM A1011/A1011M - Standard Specification for Steel, Sheet and Strip, Hot-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, and Ultra-High Strength

ASTM B6 - Standard Specification for Zinc

ASTM B201 - Standard Practice for Testing Chromate Coatings on Zinc and Cadmium Surfaces

ASTM B633 - Standard Specification for Electrodeposited Coatings of Zinc on Iron and Steel

(Copies of these documents are available online at www.astm.org.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO/IEC 17025 - General Requirements for the Competence of Testing and Calibration Laboratories

(Copies of this document are available online at www.iso.org.)

SOCIETY FOR PROTECTIVE COATINGS (SSPC)

SSPC-SP 7/NACE No. 4 - Brush-Off Blast Cleaning

(Copies of this document are available online at www.sspc.org.)

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AMERICAN WELDING SOCIETY (AWS)

AWS A5.1/A5.1M - Specification for Carbon Steel Electrodes for Shielded Metal Arc Welding

(Copies of this document are available online at www.aws.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Materials. Materials not covered by this specification shall be high grade and free from defects. Aluminum, aluminum alloys, cast iron, or malleable iron shall not be used in any part of the flame arrester assembly.

3.3 Design. The flame arrester assembly shall consist of a watertight casing, the arresting media that shall prevent the passage of a flame from one side of the media to the other side when tested as specified herein, and cleanable high velocity impingement type air filters.

3.3.1 Shock. The design of the complete flame arrester assembly shall be such that it is capable of passing the high-impact shock tests in accordance with MIL-DTL-901 for grade A, class I equipment.

3.3.2 Sizes. Sizes of flame arrester assemblies shall be limited to those in accordance with Drawing 804-1231181. The physical dimensions and tolerances of the flame arrester assembly shall be in accordance with Drawing 804-1231181 for the sizes specified.

3.3.3 Casing. The flame arrester assembly casing shall be made of galvanized steel in accordance with ASTM A653/A653M with commercial thickness zinc coating or black steel in accordance with ASTM A1011/A1011M.

3.3.3.1 Galvanization. Casings and other parts made of black steel shall be galvanized by the hot-dip process in accordance with ASTM A123/A123M with the spelter in accordance with grade LME of ASTM B6, or by electroplating with zinc in accordance with service condition 3, finish type II of ASTM B633 followed by a chromate treatment free of hexavalent chromium in accordance with ASTM B201.

3.3.4 Flame arresting media. The flame arresting media shall be constructed of galvanized steel or bronze and shall be permanently secured in the casing in a manner to prevent passing of a flame between the media and the casing.

3.3.5 Air filters. Air filters shall be provided in the air entering side of the casing to protect the arresting media from dust. The air filters shall be in accordance with MIL-PRF-16552. Means, such as flat springs, shall be provided to hold the air filters firmly within the casing.

3.3.6 Access cover. A watertight access cover shall be provided in the casing for servicing the air filters. The access cover gasket shall be of cork, neoprene, or a three-ply butyl rubber tape, fabric reinforced and pressure sensitive on both sides.

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3.3.7 Pressure loss (resistance to airflow). The resistance to airflow through the flame arrester, with air filters removed, shall not exceed the values specified in table I.

TABLE I. Pressure loss at face velocity.

Face velocity (feet per minute)	Pressure loss (inches [water gage])
400	0.27
600	0.60
800	1.10

3.4 Welding. The surfaces of all parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. All welding shall be at least equal to that in accordance with S9074-AR-GIB-010/278. Where zinc-coated steel is used for fabricating parts, the metallic zinc shall be removed from all points and surfaces on which welds are to be deposited and for a distance of ½ inch from the expected toes of the welds. In areas where the metallic zinc cannot be removed and it is necessary to weld over the zinc-coated surface, an electrode conforming to AWS A5.1/A5.1M, type E6010 shall be used.

3.5 Finish. Surfaces to be painted shall be cleaned to remove dirt, grease, corrosion products, and other surface contaminants prior to application of primer coating in accordance with SSPC-SP 7/NACE No. 4. Unless otherwise specified herein, all zinc-coated surfaces, including welds and areas damaged by welding, grinding, shearing, punching, or any fabricating process, shall receive a thorough wash and a hot phosphate acid treatment prior to applying two coats of primer in accordance with MIL-DTL-24441, formula 150. Flame arresting media and air filters shall not be painted.

3.6 Fasteners. All bolts, nuts, screws, and other such fasteners that may be used shall be of a corrosion-resisting material suitable for the intended application and shall be of a recognized commercial standard shape, size, length, and tolerances. Unless otherwise specified (see 6.2), bolts, nuts, and machine screws shall have unified-coarse-thread series threads.

3.7 Identification plates. Each flame arrester assembly shall be provided with an identification plate not to exceed 3 by 3 inches. Identification plates and other designating marking plates may be etched, engraved, or cast using either Monel (or equal), brass, corrosion-resisting steel, or aluminum alloy or may be of an anodized aluminum photographic process. These plates shall be secured to the outside of the casing in such a manner as not to destroy the watertight integrity of the casing. The identification plate shall contain the following information:

a. “Flame Arrester, Ventilation-Exhaust”
b. Size
c. National stock number (NSN)
d. Component identification number (CID)
e. Contract number
f. Manufacturer’s name
g. Manufacturer’s address

3.7.1 Direction of airflow. Each flame arrester assembly shall be provided with an airflow indication in accordance with Drawing 804-1231181.

3.7.2 Aluminum photographic process. Aluminum anodized plates shall have the photosensitive silver compounds imbedded within the oxide layer. After photographic processing, the image shall be sealed in the oxide layer by chemical treatment of the layer. Stamping of additional marking information will not normally be permitted on metal plates.

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3.8 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed on a flame arrester when a first article sample is required (see 3.1). Unless otherwise specified (see 6.2), one sample of each new design or a different size not covered by a previous first article inspection (see 6.3) shall be submitted for first article inspection. First article inspection shall consist of the examination specified in 4.5 and the tests specified in 4.6.1 through 4.6.4.

4.2.1 First article extension. Where an identical size flame arrester has successfully passed the first article inspection within a period of not more than 3 years of the date of the contract or order, the first article inspection may be extended by the command or agency concerned.

4.3 Conformance inspection.

4.3.1 Lot. All flame arresters of the same size offered for delivery at one time shall be considered a lot.

4.3.2 Sampling for examination. A random sample of flame arresters shall be selected from each lot in accordance with table II for the examination specified in 4.5 at the Acceptance Quality Limit (AQL) specified (see 6.2).

TABLE II. Sampling for examination.

Number of arresters in lot	Number of arresters to be examined
Up to 8	All
9 to 15	7
16 to 25	10
26 to 65	15
66 to 110	20
111 to 180	25
181 to 300	35

4.3.3 Sampling for tests. A sample flame arrester shall be selected from each lot for the tests specified in 4.6.2 through 4.6.4.

4.4 Periodic tests. Periodic tests shall be conducted to ensure continuing satisfactory operation of identical flame arresters. The tests shall be conducted under any contract or order for flame arresters of existing design when the contract or order is dated 3 years or more subsequent to the date of the previous first article examination and tests of an identical flame arrester. Periodic tests shall be conducted on each size flame arrester as specified (see 6.2) and shall consist of the examination and tests specified in 4.2.

4.5 Examination. Flame arresters shall be examined and measured to verify compliance with the requirements of this specification not involving tests. Examination shall be conducted to identify defects as specified in table III. One or more defects in any flame arrester shall be cause for rejection of the flame arrester, and a number of defective units that exceeds the acceptance number for the sample shall be cause for rejection of the lot.

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TABLE III. Classification of defects.

Categories	Defects
Critical	None defined.
Major	
101	Size not as specified.
102	Incomplete; component parts missing.
103	Evidence of unauthorized material used.
104	Casing or enclosure material thickness or inside and outside dimensions nonconforming.
105	Welding (as applicable) incomplete, not free of cracks, non-fusion, heavy porosity, heavy undercut, slag inclusions, or not smooth as specified.
106	Casing or enclosure drilled or punched holes nonconforming; holes missing or location and size not as required.
107	Provisions not provided to hold filter assembly springs in place; springs nonconforming or not located as required.
108	Casing or enclosure hardware (for example, wingnuts, lockwashers, bolts, and nuts) not provided; size and length nonconforming.
109	Casing surface not free of cracks, splits, tears, burrs, and sharp edges, or not clean prior to galvanizing or electroplating with zinc as specified.
110	Casing and arresting media coating incomplete; areas uncoated (bare spots) or not free of sharp projections.
111	Air filter missing, defective, or not as specified.
112	Access cover and gasket missing, defective, or not as specified.
113	Painting (as applicable) nonconforming.
114	Marking, manufacturer's name or trademark, or direction of airflow arrows missing; markings not permanent, incorrect, or illegible.

4.6 Tests.

4.6.1 Shock test. The flame arrester shall be shock tested on the lightweight or medium weight shock machine and the test shall be in accordance with shock test type A of MIL-DTL-901. The lightweight or medium weight shock machine shall be selected based on the weight and fixture limitations specified in MIL-DTL-901. The shock test shall be conducted prior to the tests specified in 4.6.3 and 4.6.4. Evidence of fragmentation or missile effect of parts or deformation of parts that will permit passage of a flame or destroy the watertight integrity of the casing shall be cause for rejection.

4.6.2 Pressure loss test. The pressure drop in the air stream across the flame arrester, with the air filter removed, shall be determined at 400-, 600-, and 800-foot per minute air velocities. The pressure drop of the flame arrester with the air filters removed shall not exceed that specified in 3.3.7.

4.6.3 Flame resistance test. The flame resistance test shall determine that the flame arrester will prevent the passage of flame from a combustible mixture of gasoline vapor and air on one side of the arrester to a similar mixture on the other side of the arrester when the arrester is installed in a section of ventilation duct. The air filters shall be removed from the casing for this test.

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4.6.3.1 Test arrangement. The test arrangement shall be as shown on figure 1 and as follows:

a. Transitions shall be symmetrical with a maximum included angle of 15 degrees.

b. The arrester shall be installed with the direction of airflow as shown on figure 1.

c. Flanged connections shall be used.

d. Duct and transition pieces shall be made from steel of the thickness and dimensions specified in table IV.

e. Ends of ducts shall be fitted with loose fitting plugs that will be blown out by the force of the explosion when the mixture is ignited.

f. An orifice shall be fitted on the ignition end of the 25-foot duct. The orifice shall be such to raise the airflow pressure drop of the whole setup including the arrester to 3 inches of water.

g. White gasoline or equal shall be used having an octane rating of 55 or higher.

h. Ductwork shall be completely filled with gasoline and air in proper proportions and shall be distributed evenly within the ductwork. The arrester shall be tested at the lower explosive limit, upper explosive limit, and the midpoint of the two concentrations for the mixture.

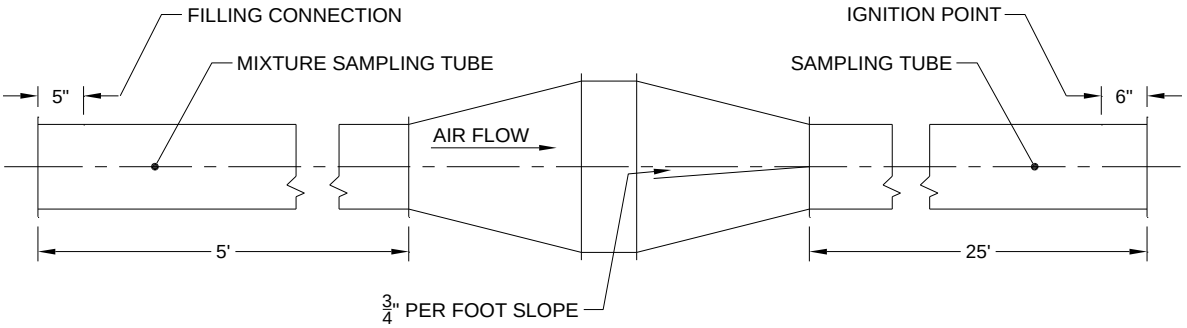


FIGURE 1. Test setup for flame arrester.

TABLE IV. Thickness and dimensions of steel for duct and transition pieces.

Size	USS gage	Duct cross section (inches)
31	26	4 by 4
32	24	5½ by 5½
33	24	7 by 7
34	22	8½ by 8½
35	22	10 by 10
36	20	13 by 13
37	18	16 by 16
38	18	21 by 21

4.6.3.2 Explosive mixture. Tests shall be conducted at three different gasoline-air vapor mixtures (lower explosive limit, upper explosive limit, and midpoint of the two) using white gasoline or equal having an octane rating of 55 or higher.

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4.6.3.3 Filling with vapor. Ends of ducts shall be fitted with loose fitting plugs that will be blown out by the force of the explosion when the mixture is ignited. These plugs shall maintain the concentration of the explosive mixture until the mixture is ignited. A hose connection shall be fitted to the 5-foot duct for filling purposes as shown on figure 1. Carbureted pure gasoline shall be fed into the test arrangement until the arrangement is uniformly filled throughout.

4.6.3.4 Ignition. After the ductwork has been filled evenly with the explosive mixture and the filling hose has been removed, the mixture shall be ignited by means of a spark plug located 6 inches from the end of the 25-foot duct as shown on figure 1.

4.6.3.5 Cause for rejection. Failure to prevent the passage of flame from one side of the arrester to the other shall be cause for rejection.

4.6.4 Watertightness test. The ends of the flame arrester assembly shall be blanked off and the casing subjected to hydrostatic testing of 10 pounds per square inch gage for a period of not less than 10 minutes. A drop in hydrostatic gage pressure in excess of 0.125 pounds per square inch in the 10-minute period shall be cause for rejection.

4.6.5 Fire testing provisions. All fire tests specified in this document shall be conducted by an independent testing laboratory that is accredited to ISO/IEC 17025 and is approved by the Technical Authority (see 6.5). Accreditation shall be obtained from a recognized accreditation body such as the American Association for Laboratory Accreditation (A2LA) (www.A2LA.org) or the International Code Council's International Accreditation Services (IAS) (www.IASonline.org). The scope of accreditation shall include specific flammability and fire tests. All other fire test provisions shall be as specified (see 6.2 and 6.4).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The arrester assembly is for shipboard applications utilizing Navy standard filters. Flame arrester sizes are in accordance with the latest revision of Drawing 804-1231181. Current stock of flame arresters may be used until depleted; future requisitions for flame arresters should refer to the current version of this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. If bolts, nuts, and machine screws should have other than unified-coarse-thread series threads (see 3.6).

c. When first article inspection is required (see 4.2).

d. AQL of 1 (see 4.3.2).

e. Requirements for periodic testing (see 4.4).

f. Additional fire testing provisions (see 4.6.5 and 6.4).

g. Packaging requirements (see 5.1).

h. When drawings delineating the equipment should be furnished. When drawings are to be furnished, drawings should be in accordance with ASME Y14.100.

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6.3 First article inspection. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Additional fire testing provisions. The Technical Authority reserves the right to witness the tests and perform any of the tests set forth herein where such testing is deemed necessary to assure compliance to prescribed requirements. A2LA (www.A2LA.org) or International Code Council's IAS (www.IASonline.org) maintain a full list of the approved laboratories that are accredited to ISO/IEC 17025 and provide details on how to obtain and maintain accreditation (see 6.2).

6.5 Technical Authority. The Technical Authority has the authority, responsibility, and accountability to establish, monitor, and approve technical standards, tools, and processes in conformance to higher authority policy, requirements, architectures, and standards. NAVSEA 05S may be contacted to determine the appropriate Technical Authority for any given requirement related to equipment and systems of U.S. Navy ships and submarines.

6.6 Subject term (key word) listing.

Duct

Ductwork

Exhaust

Fire protection

Flame

Flame mitigation

Flame travel

Flammable mixture

Ventilation

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Preparing activity:
Navy – SH
(Project 2040-2019-003)

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